Assignment: Text Emotion Classification using Word Embedding and LSTM

1. Assignment Objective

Develop an NLP-based Deep Learning application that classifies text into different emotion categories using **Word Embedding and LSTM**.

2. Requirements

You must:

1. Select or use a suitable text emotion dataset.
2. Prepare and preprocess the text data.
3. Split the dataset into **Training** and **Testing** sets.
4. Perform text tokenization and sequence padding.
5. Use a **Word Embedding** layer to represent words as numerical vectors.
6. Build and train an **LSTM-based Deep Learning model**.
7. Evaluate the model using suitable evaluation metrics.
8. Save the trained model.
9. Develop a simple **Streamlit UI** for emotion prediction.
10. Test the application using new text inputs.

3. Emotion Classes

The project should classify text into at least **4 emotion categories**.

Examples:

- Joy
- Sadness
- Anger
- Fear
- Love
- Surprise

You may use different emotion categories depending on their selected dataset.

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4. Model Requirements

The Deep Learning model must include:

- Word Embedding Layer
- LSTM Layer
- Dense Output Layer
- Appropriate activation function for multi-class classification

5. Streamlit Application

The application should allow the user to:

- Enter a text sentence
- Click a **Predict** button
- Display the predicted emotion
- Display prediction confidence if possible

6. Submission Requirements

Submit the following:

- Python source code
- Trained model file
- Dataset or dataset source
- requirements.txt
- Streamlit application
- Short project report

7. Suggested Project Title

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